Multi Agent Euclidian ~~MDP~~ ??

Multiplayer stochastic game.

A family of Euclidean M~~D~~P ??

$\bar{\mathcal{A}} = \{\mathcal{A}_1, \dots, \mathcal{A}_n\}$ where

$$\mathcal{A}_i = (S_i, s_0^i, Act_i, T_i, C_i, G_i)$$

with

- $S_i \subseteq \mathbb{R}^{k_i}$ $\quad k_i \in \mathbb{N}$ (bounded, closed)
- Act_i $\quad$ finite set of actions.
- $T_i : \underbrace{\bigtimes_i S_i}_{\text{Global state}} \times Act_i \to \underbrace{(S_i \to \mathbb{R}_{\geq 0})}_{\text{density}}$ $\quad \times\{1..n\}$ whose turn next
- $C_i : \bigtimes_i S_i \times Act_i \to \mathbb{R}$ (cost)
- $G_i \subseteq S_i$ (goal-states)

(Mixed) Strategy σ_i for $\mathcal{A}_i$:

$$\sigma_i : \bigtimes_i S_i \to \underbrace{(Act_i \to [0,1])}_{\text{prob distr}}.$$

A family of Euclidean MDP

$\bar{\mathcal{A}} = \{\mathcal{A}_1, \dots, \mathcal{A}_n\}$ where

$\mathcal{A}_i = (S_i, s_0^i, Act_i, T_i, C_i, G_i)$

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- $S_i \subseteq \mathbb{R}^{k_i}$ $\quad k_i \in \mathbb{N}$ (bounded, closed)
- Act_i $\quad$ finite set of actions.
- $T_i : \underbrace{\bigtimes_i S_i}_{\text{Global state}} \times Act_i \to \underbrace{(S_i \to \mathbb{R}_{\geq 0})}_{\text{density}}$ $\quad \times \{1..n\}$ $\quad$ (where turn next)
- $C_i : \bigtimes_i S_i \times Act_i \to \mathbb{R}$ (cost)
- $G_i \subseteq S_i$ (goal-states)

Strategy σ_i for $\mathcal{A}_i$:

$\sigma_i : S_i \to Act_i$ $\quad$ Can be made prob & non-det

Strategy profile

$\bar{\sigma} = (\sigma_1, \dots, \sigma_n)$

Global configuration $C \in \underbrace{\bigtimes_i S_i}_{\text{state for each component}} \times \underbrace{\{1, \dots, n\}}_{\text{whose turn}}$

Initial configuration $C_0 = ((s_0^1, \dots, s_0^n), 1)$

Given strategy profile $\bar{\sigma} = (\sigma_1, \dots, \sigma_n)$

$$\gamma = C_0 \to C_1 \to C_2 \to \dots$$

is a valid run if for all i.

- $C_i = (\bar{s}_i, j_i)$
- $T_{j_i}(\bar{s}_i, \sigma_{j_i}(\bar{s}_i))(\bar{s}_{i+1}, j_{i+1}) > 0$
- $C_{i+1} = (\bar{s}_{i+1}, j_{i+1})$

$\bar{\sigma} = (\sigma_1 \ldots \sigma_n)$ $\qquad \sigma_i : \bigtimes_i St_i \to Act_i$

Configuration $\quad C \in \bigtimes_i St_i$

$C_0 \xrightarrow[i_0]{} C_1 \xrightarrow[i_1]{} C_2 \xrightarrow[i_2]{} \quad \to C_n \to$

run following $\bar{\sigma}$

$(C_0, i_0) \to (C_1, i_1) \to \ldots \to (C_n, i_n) \to$

Global Configuration $\quad C \in \bigtimes_i St_i \times \{1 \ldots n\}$

Global Scheduler $\gamma \in \bigtimes_i St_i \times \{1 \ldots n\} \to \{1 \ldots n\}$

$(\{1 \ldots n\} \to [0,1])$

$\bar{\sigma} = (\sigma_1 \ldots \sigma_n)$ $\qquad$ $\bar{s} = (s_1 \ldots s_n)$

$$((s_1, \ldots, s_n), i) \longrightarrow ((s_1', \ldots s_n'), i')$$

$$T(s_i, \sigma_i(\bar{s}))(s_i') > 0 \quad \wedge$$

$$NOP_i(s_j, s_j') \quad \wedge$$

$$\gamma(\bar{s}, i)(i') > 0$$

Periodic Switched Control

$\mathcal{A}_i$ Modes $\mathcal{M}_i = \{m_1 \ldots m_k\}$

Period τ_i

$Act_i \subseteq \mathcal{M} \times \mathcal{M}$

States $S_i \subseteq \mathbb{R}^k$

Trajectory $T_i(m_j, s_0, t) \in S_i$

state after t tu starting in s_0 using mode m_j.

Strategy $\sigma_i : \bigtimes_i S_i \to Act_i$

Multiagent System $\bar{\mathcal{A}} = (\mathcal{A}_1 \ldots \mathcal{A}_n)$

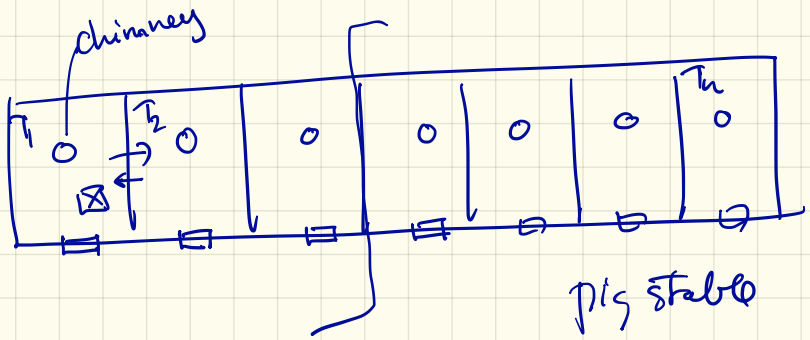
Strategy profile $\bar{\sigma} = (\sigma_1 \ldots \sigma_n)$

Configuration (C, i) $\quad C \subseteq \bigtimes_i S_i \times \bigtimes_i \mathcal{M}_i \quad i \in \{1 \ldots n\}$

Run

$(\bar{s}, \bar{m}, i) \longrightarrow (C_1, i_1) \longrightarrow (C_2, i_2) \ldots$

- $\mathcal{A}_i$ change mode m_i' using some action
- Evolve according to $\bar{m}[m_i'/m_i]$


chimney
T_1
T_2
T_n
pig stable